

MINI-PROJECT REPORT

UNIT 1: INTRODUCTION TO NETWORKING AND DATA COMMUNICATION

Mini-Project Title

Building a LAN in Cisco Packet Tracer

Submitted By

Name: Shreyas Babasaheb Kote

PRN: 2125UCSM1022

Department: Cybersecurity

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1. OBJECTIVE

The objective of this mini-project is to design and configure a Local Area Network (LAN) using Cisco Packet Tracer. The network consists of one router, two switches, and four PCs.

The project demonstrates the configuration of static IP addresses, subnet masks, default gateways, router interfaces, and basic network connectivity. Successful communication between the devices is verified using the ping command.

2. SOFTWARE USED

Software: Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool used to create and configure virtual networking environments. It allows network devices such as routers, switches, and computers to be connected and configured without requiring physical hardware.

3. NETWORK TOPOLOGY

The LAN was designed using the following devices:

- 1 Router
- 2 Switches
- 4 PCs
- Ethernet connections

The router connects the two switches, while each switch connects two PCs.

The network topology is represented as:

```

ROUTER
  /   \
G0/0   G0/1
  /     \
SWITCH 1 SWITCH 2
  /   \   /   \
PC1   PC2 PC3   PC4

```

Figure 1: LAN Topology

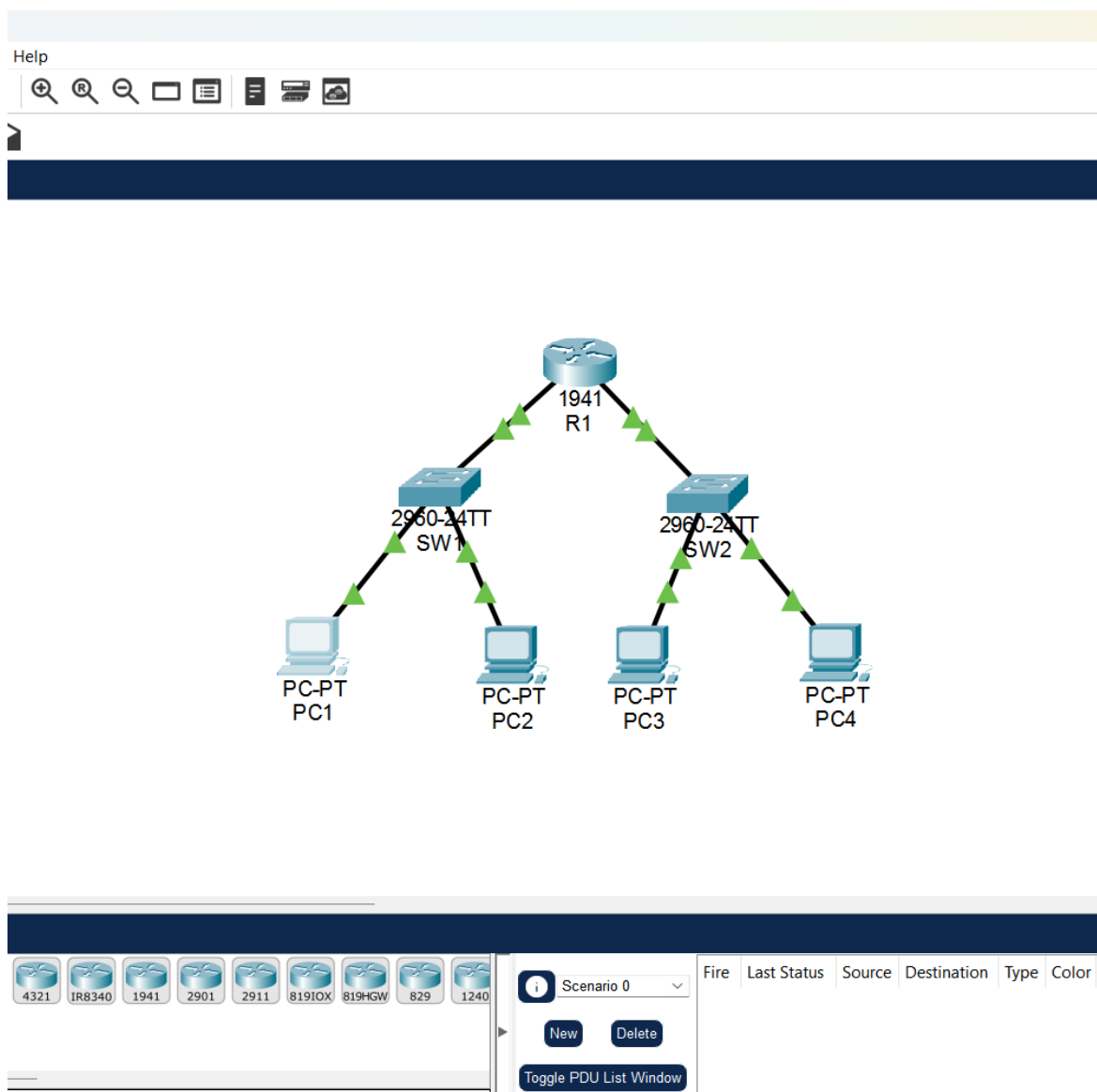


Figure 1: Network topology created using Cisco Packet Tracer.

4. IP ADDRESSING

Static IP addressing was used for all four PCs.

Two separate networks were configured:

- **192.168.1.0/24** for PC1 and PC2
- **192.168.2.0/24** for PC3 and PC4

The router acts as the gateway between the two networks.

IP Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.11	255.255.255.0	192.168.1.1
PC3	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.11	255.255.255.0	192.168.2.1
Router G0/0	192.168.1.1	255.255.255.0	—
Router G0/1	192.168.2.1	255.255.255.0	—

5. PROCEDURE

Step 1: Creating the Network Topology

Cisco Packet Tracer was opened and the required network devices were added to the workspace.

The following devices were used:

- One router
- Two switches

- Four PCs

The PCs were connected to the switches using Ethernet connections. Switch 1 and Switch 2 were then connected to the router through its GigabitEthernet interfaces.

The final topology consisted of two LAN segments connected through a router.

Step 2: Configuring the PCs

Static IP addresses were assigned to each PC.

To configure a PC, the following steps were followed:

PC → Desktop → IP Configuration → Static

The IP address, subnet mask, and default gateway were entered according to the IP addressing table.

For example, PC1 was configured as follows:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC2, PC3, and PC4 were configured similarly with their respective IP addresses and gateways.

Step 3: Configuring the Router

The router was configured using the Command Line Interface (CLI).

The first GigabitEthernet interface was configured for the 192.168.1.0/24 network, while the second interface was configured for the 192.168.2.0/24 network.

The following commands were used:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

The no shutdown command was used to activate the router interfaces.

Step 4: Verifying Router Interfaces

The router configuration was verified using the following command:

show ip interface brief

The GigabitEthernet interfaces were displayed as **up/up**, indicating that the interfaces were active and operational.

Figure 2: Router Interface Verification

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up          up
GigabitEthernet0/1       192.168.2.1     YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

Figure 2: Verification of router interfaces and IP addresses.

6. CONNECTIVITY TESTING

After configuring the PCs and router, network connectivity was tested using the ping command.

First, communication between PCs on the same network was tested.

For example:

ping 192.168.1.11

was used from PC1 to test connectivity with PC2.

Communication between the two different networks was then tested.

From PC1, the following command was used:

ping 192.168.2.10

This tests communication between PC1 and PC3 through the router.

The network path is:

PC1 → Switch 1 → Router → Switch 2 → PC3

A successful response confirms that the two networks can communicate through the router.

7. PING RESULT

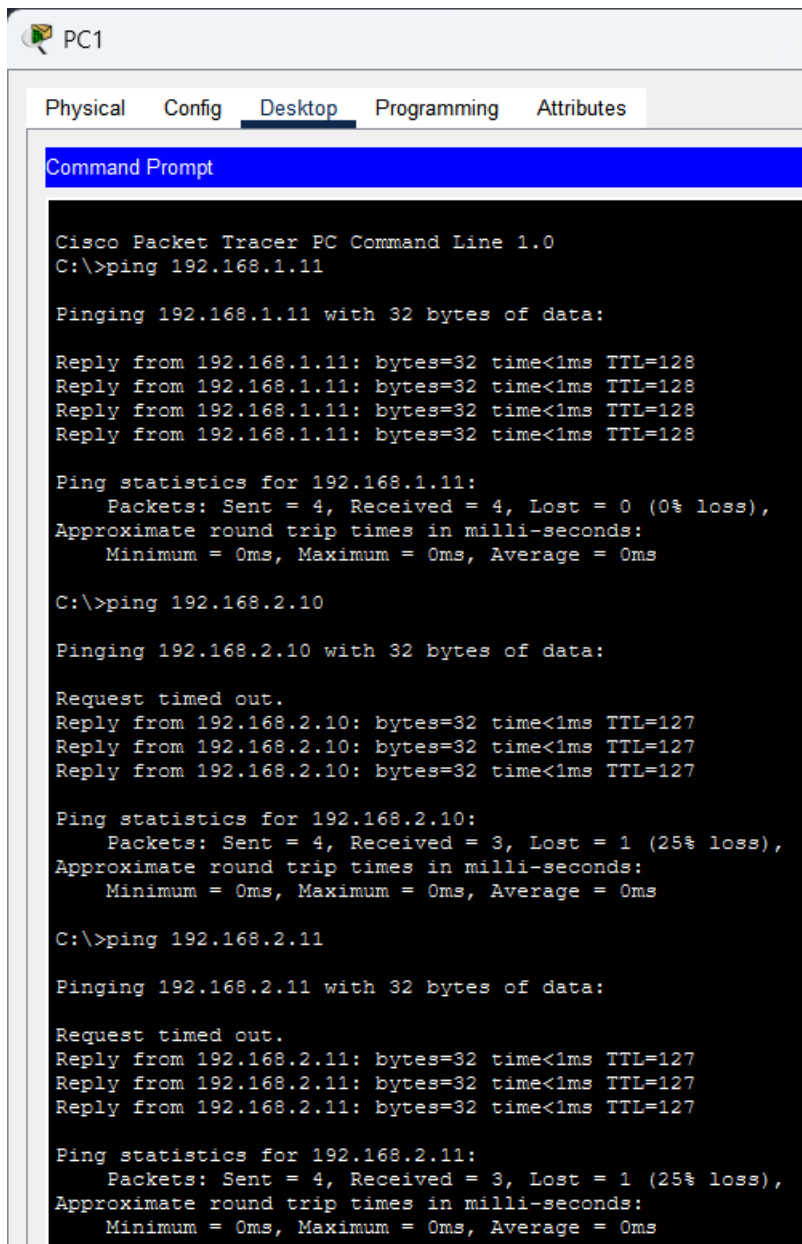
The ping test was successfully completed and responses were received from the destination PC.

The result showed:

Packets: Sent = 4, Received = 4, Lost = 0

This indicates that all four ICMP packets were successfully delivered and received.

Figure 3: Successful Ping Test



The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A Command Prompt window is open, displaying the results of a ping test from PC1 to 192.168.1.11. The output shows four successful replies with 0% loss.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 3: Successful connectivity test between PC1 and PC3.

8. NETWORK CONFIGURATION SUMMARY

The final network configuration can be summarized as follows:

Router G0/0 → 192.168.1.1

Router G0/1 → 192.168.2.1

PC1 → 192.168.1.10

PC2 → 192.168.1.11

PC3 → 192.168.2.10

PC4 → 192.168.2.11

Both router interfaces were active, and the PCs were configured with appropriate default gateways.

9. RESULT

The Local Area Network was successfully designed and configured using Cisco Packet Tracer.

Static IP addresses were assigned to all four PCs, and the router interfaces were configured with appropriate IP addresses. The router successfully connected the two different network segments.

Connectivity between devices was verified using the ping command, and successful responses were received.

10. CONCLUSION

This mini-project provided practical experience in designing and configuring a basic Local Area Network using Cisco Packet Tracer.

The project demonstrated the use of routers and switches, static IP addressing, subnet masks, default gateways, router interface configuration, and ICMP-based connectivity testing.

The successful ping tests confirmed that communication was established between devices on the same network as well as between devices on different networks through the router.
